DETF Litepaper

Reserve-Backed Onchain Shares, Nested Mark Integrity, and Bond-Ledger Rewards

IndexedEx Research

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Contents

Abstract

A **DETF** (**D**ecentralized **ETF**) is an onchain product pattern: one ERC-20 share over a real multi-asset reserve, with mint, burn, and synthetic valuation derived from that same reserve—not from an admin spreadsheet.

Instances deploy **inert**, go **live** on first successful bond, and choose deploy-time **Policy** (synthetic price gates primary mint and burn) or **Open** (no price gates when live). Nested Standard Exchange (SE) legs can keep marks fair under underlying demand when rate providers are on. Residual lag under rates-off is real, but it is not free arbitrage once fees bind.

This paper formalizes synthetic price, Policy and Open gates, **natural supply expansion** (Policy and mint-rich only; bond ledger only), and **protocol compound** (protocol bond rewards sink into protocol-owned reserve BPT). Hermetic Single SE DETF scenarios and prior Uni V2 SE rate-provider matrices supply measured support.

This is not a prospectus and not a yield forecast. We do not claim mainnet APY, peg guarantees, registered-fund status, or expansion yields for free unlocked holders.

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1. Introduction

ETF-shaped demand wants one transferable share over a basket. Onchain attempts often fail in predictable ways: discretionary managers, off-pool "NAV" that disagrees with the market, opaque rebalancers, or nested vault legs that freeze mid prices while underlyings move.

A DETF answers with:

- **Reserve-pool pricing** as the single engine for primary mint, burn, and synthetic valuation
- **Immutable, unowned instances** after deploy for normal operation
- An explicit **inert -> live** lifecycle via bonding
- Nested SE vaults plus Balancer rate providers for mark integrity on vault-share legs
- Bond-ledger rewards for capital seigniorage and (on Policy units) natural expansion
- Protocol compound that sinks protocol rewards into reserve BPT

Merits are stated as **properties**, not performance promises. Measured results come from production-first hermetic research—not mocks of the subject under test.

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2. Background: nested Standard Exchange

Standard Exchange vaults wrap protocol liquidity (for example Uni V2 LP) behind deposit and redeem share interfaces. When a Balancer pool holds SE shares as legs, two deploy intents matter:

Intent	Effect under Uni demand
Rates on (R+)	Pool mids re-mark with SE redeem rates; residual approx 0
Rates off (R–)	Mids can freeze while rates track Uni; residual grows

Intent	Effect under Uni demand

Residual (research fairness metric):

```
residual approx  mid_index * rate_index - 1
```

Residual measures mark lag. It is **not** profit. A closer must clear Balancer swap fee, SE usage fee, impact, and dust. In the SE research package, the Balancer constant-product fee was **5%**—a hard economic filter.

Under baseline Uni-only demand: R+ residual approx 0; R– residual ~+/-0.24%. At extreme volume (size mul=25, 48 steps): R– residual ~10–12%, and Mode C fills appear once residual exceeds fee scale.

We cite industry work on LVR and adverse selection for *why* fees and inventory matter. We do not re-derive that theory here.

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3. How a DETF works

3.1 Roles and reserve

The DETF diamond **is** the share ERC-20. A typical Single Standard Exchange composition:

- One **SE vault** and its **vault share**
- A Balancer V3 **weighted reserve** including the DETF self-leg and external legs
- A **bond NFT vault** for locked principal and reward accounting
- Optional **rebasing claim** on protocol-owned reserve BPT

Production DETF code talks to Standard Exchange interfaces, share ERC-20s, and Balancer—not venue brands baked into the product definition.

3.2 Synthetic price

```
p_syn = (rate-scaled claim of owned reserve BPT on pool balances)
        / (DETF totalSupply)
```

Bond-vault BPT is included when peer families do. The abstract Policy peg narrative is **1e18**. **All mint and burn threshold gates use synthetic price**—never spot alone.

3.3 Lifecycle: inert -> live

State	User primary mint	Protocol depth
Inert	Blocked	Not live
Live (after first successful bond)	Subject to Policy or Open	Bond principal in protocol accounting

First bond is **synthetically ungated**. Hermetic research confirms: inert deploy blocks mint; warp while inert does not enable mint; first bond takes the unit live and sets the reserve pool.

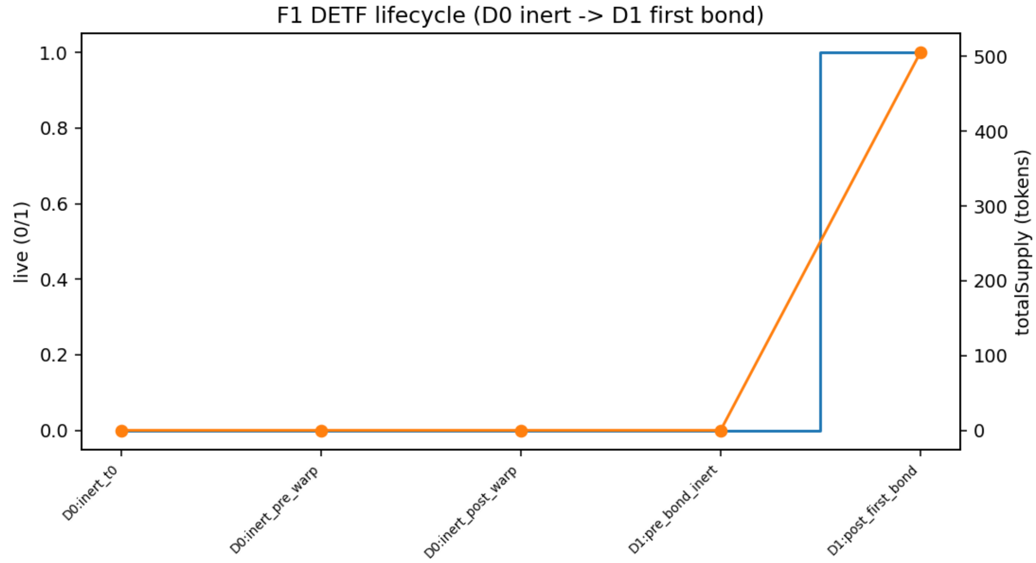

Figure 1: Lifecycle: inert Policy deploy, first bond, live reserve.

3.4 Policy vs Open

	Policy (default)	**Open**
Mint when live	`p_syn > mintThreshold`	Always (route rules still apply)
Burn when live	`p_syn < burnThreshold`	Always
Defaults if args are zero	mint 1.05e18 / burn 0.95e18	Same stored defaults; gates ignore them
Natural expansion	When mint-rich + time	**Never**
Zero args imply Open?	**No**	Mode must be explicit at deploy

Hermetic Single SE results: after first bond, synthetic often sits burn-side of peg under default thresholds, so Policy mint is blocked until a rich path is established; Open mint and burn execute when live without synthetic gates.

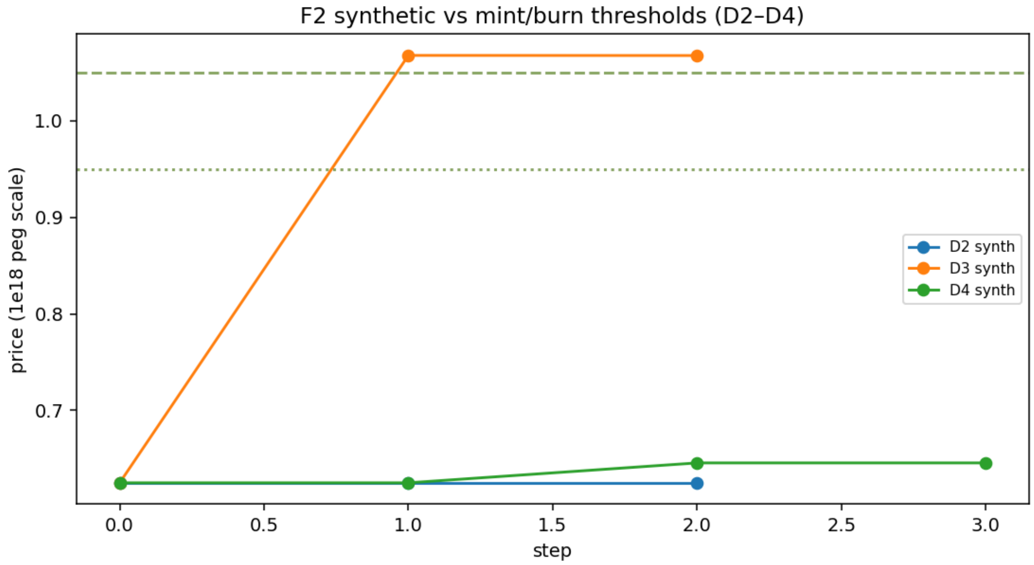

Figure 2: Synthetic price versus default mint and burn bands under Policy.

3.5 Immutability

True DETF instances are **immutable and unowned** after deploy for normal operation: no instance owner, no discretionary diamond cut, no admin pause as the product model. Flawed config means abandon and redeploy—not silent rewrites of a live instance.

3.6 Routes and preview honesty

Preferred routes are closed-form **vault shares <-> DETF**. Rate-asset-as-mint-input is family-scoped only. Cross-share routes on the DETF surface are out of scope; use Balancer or the Standard Exchange router on the reserve pool.

On the measured capital mint path, **preview amount equaled execution amount exactly** (difference zero).

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4. Rewards: three paths, three labels

Keep capital seigniorage, natural expansion, and protocol compound separate.

4.1 Capital seigniorage

External capital (SE shares) mints DETF when gates allow. Usage fees and inventory splits feed the **bond reward ledger** as the family defines. This is **not** expansion.

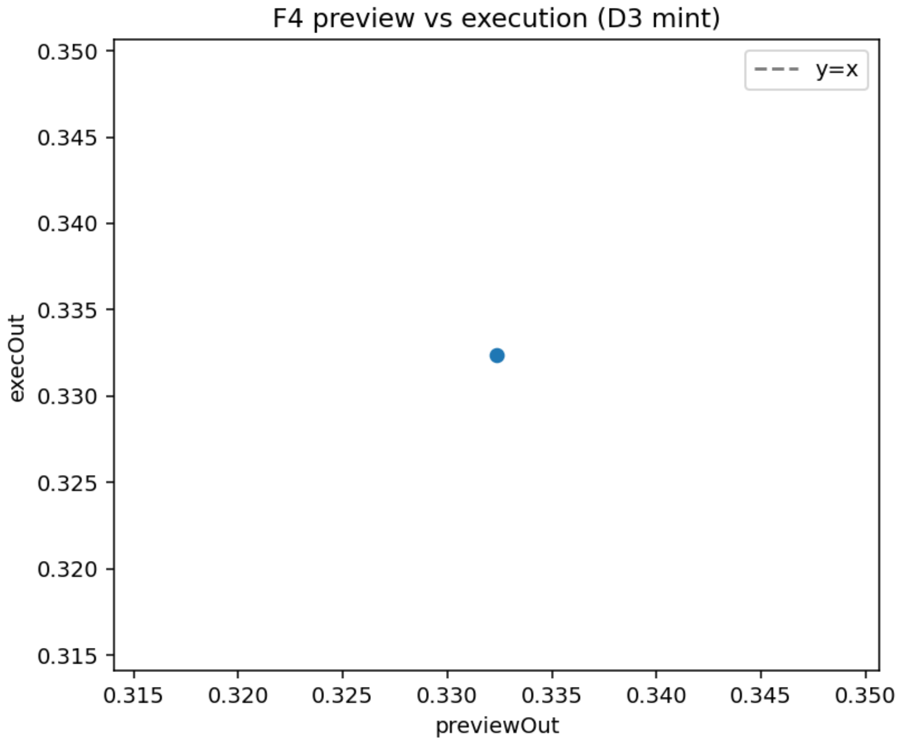

Figure 3: Closed-form capital mint: preview versus execution.

4.2 Natural supply expansion

When **Policy**, **live**, and **`p_syn > mintThreshold`**, free DETF may mint over time into the bond reward vault (premium-closure formula; deploy-time rate and catch-up caps). Distribution follows **bond effective shares**. Free unlocked DETF holders receive **none**. **Open never expands.** Inert and non-rich units do not expand.

Hermetic results: Open supply and pending rewards are unchanged over long warps; Policy expands when rich after time and a product touch; free-only holders do not receive expansion airdrops.

4.3 Protocol compound

Protocol-owned NFT free DETF rewards **auto-sink** into more **protocol-owned reserve BPT** via a single-sided DETF join (best-effort; also callable via public `compoundProtocolRewards`). Users still **claim free DETF** on their own bonds. Claim redemption **can** improve when protocol BPT rises—this is not a coupon.

Hermetic D9 shows protocol NFT BPT principal rising after compound.

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5. Merits as properties

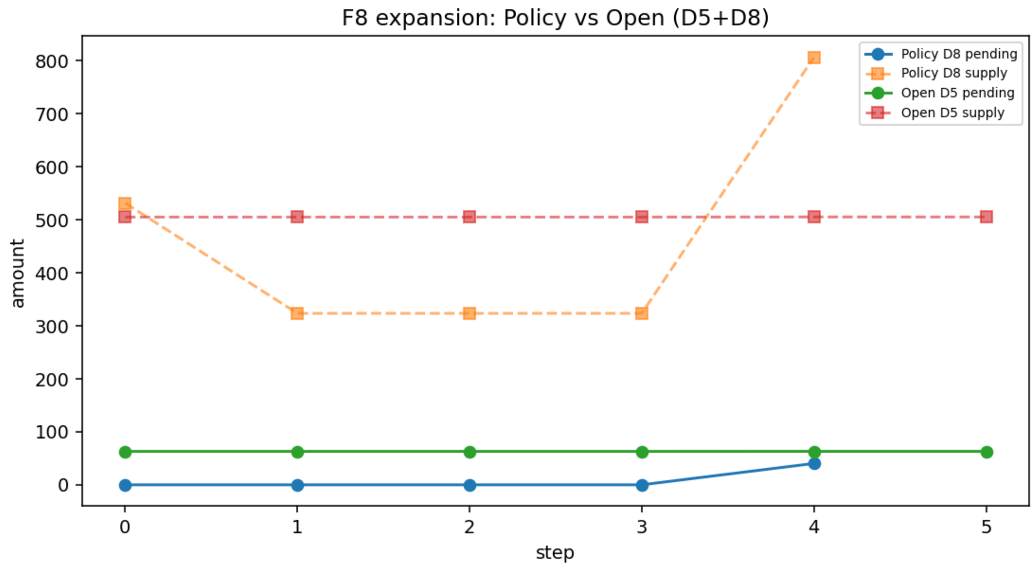

Figure 4: Natural expansion under Policy when rich versus Open control.

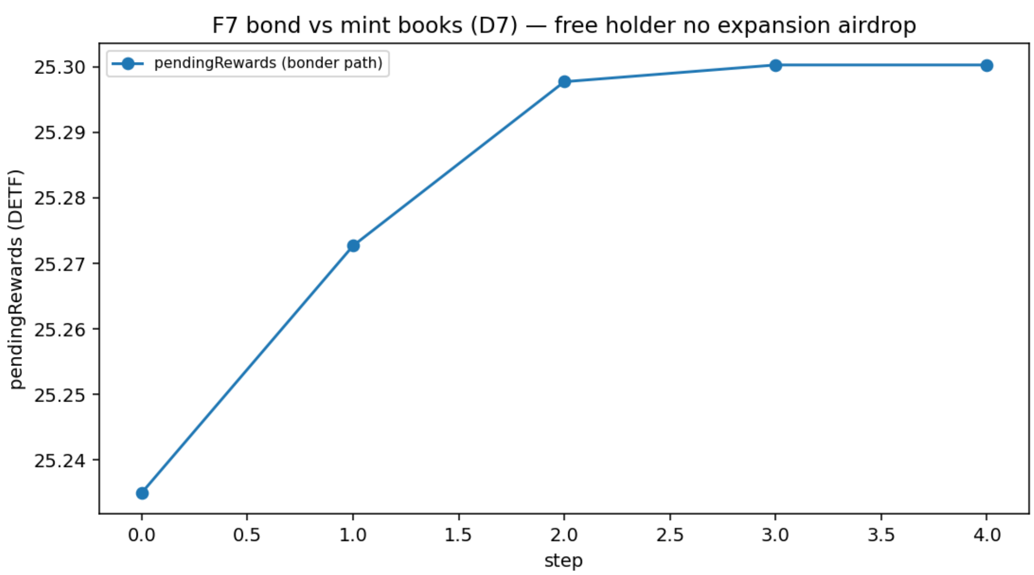

Figure 5: Bond reward path versus free unlocked DETF holder.

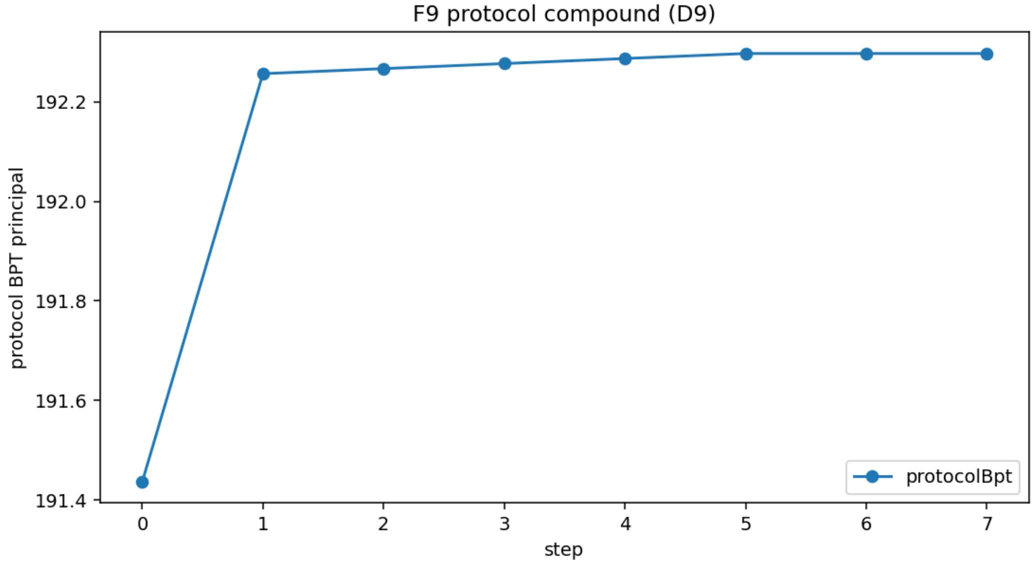

Figure 6: Protocol compound raises protocol-owned BPT principal.

Property	Evidence
ETF-shaped basket exposure without a discretionary PM	Design
Mint, burn, and synthetic share one pricing engine	Design + measured DETF
Clear inert/live; bonding builds protocol-owned depth	Measured DETF
Explicit monetary policy (Policy vs Open) at deploy	Product law + measured
No admin rewrite of a live instance as product model	Product law
Nested SE legs with rates on keep residual approx 0 under Uni demand	Measured SE
Lag invites reprice only after the fee stack clears	Measured SE
Liquid mint vs bond reward books	Measured DETF
Natural expansion when rich (Policy only)	Measured DETF
Protocol compound deepens protocol BPT	Measured DETF
Keeper-free accrual and compound	Product law + code
Composable package types for different baskets	Design taxonomy
Platform: many DETFs; optional protocol fee path	Product hierarchy

6. Nested mark integrity (measured SE)

Headline. With rate providers on, Uni demand re-marks Balancer mids so midxrate residual approx 0. With rates off, residual scales with Uni stress. Residual is not free arb: under a 5% research pool fee, modest residual never filled; extreme residual did.

Tier	R+ residual	R– residual	Mode C R– fills
Baseline	approx 0	~+/-0.24%	0
mul10	approx 0	~+/-2.4%	0 (fee-drowned)
mul25 x 48 steps	approx 0	~+/-10–12%	Fills from ~step 22

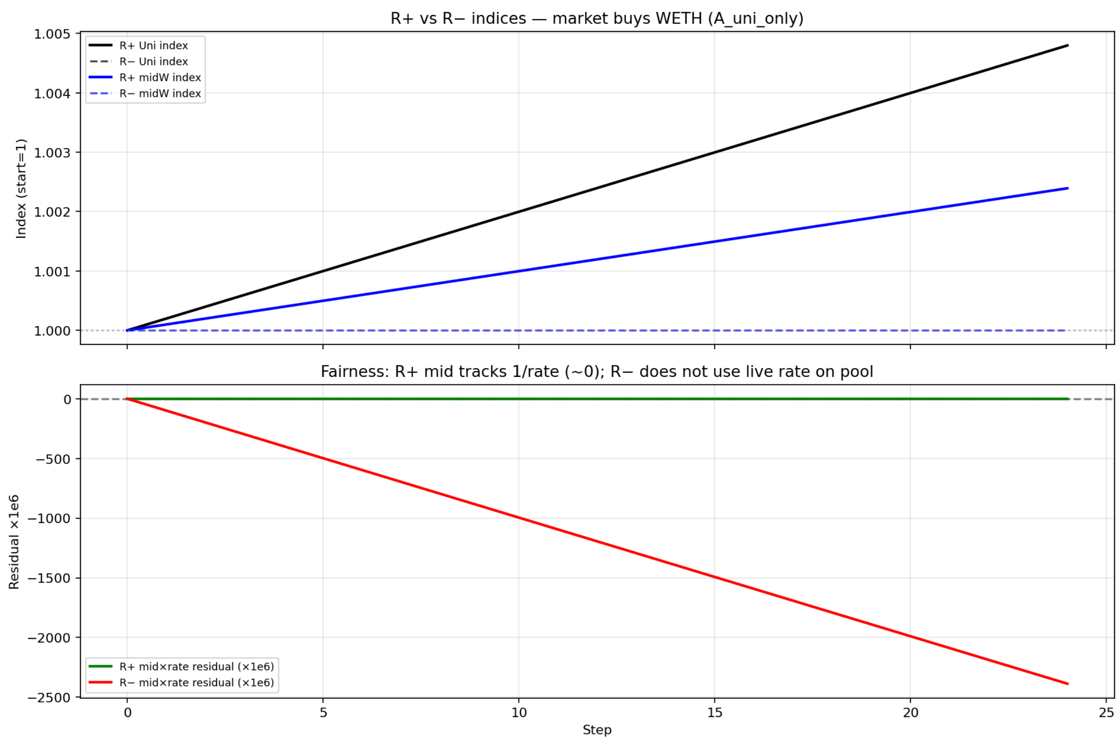

Figure 7: R+ versus R– residual under Uni-only demand (baseline).

Transferability. Hermetic only. The research Balancer fee is not every production fee tier. Cite mechanism, not universal fill rates.

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7. DETF scenarios (measured Single SE + Uni V2)

Harness. Production-first CREATE3 and registry DFPkg path; Uni V2 WETH/USDC SE; Foundry default profile. One forge script per scenario D0–D9.

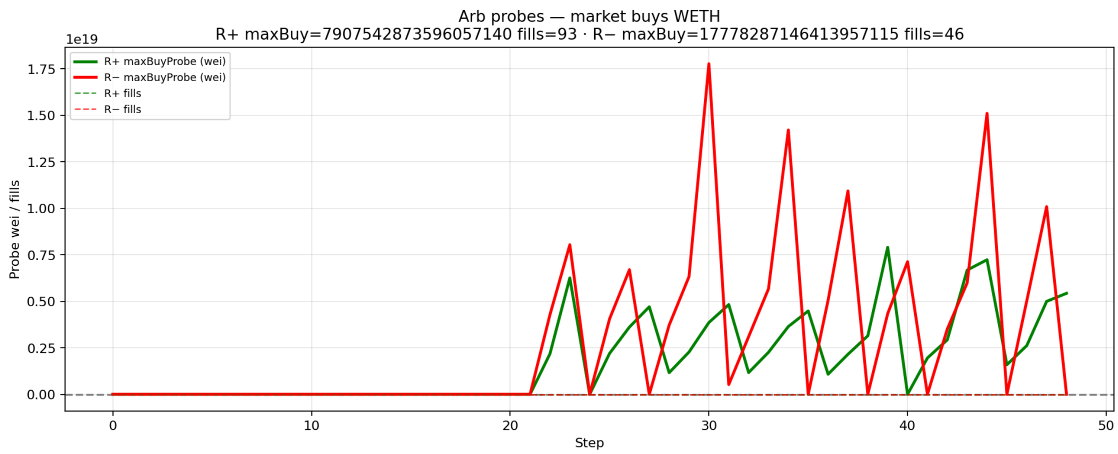
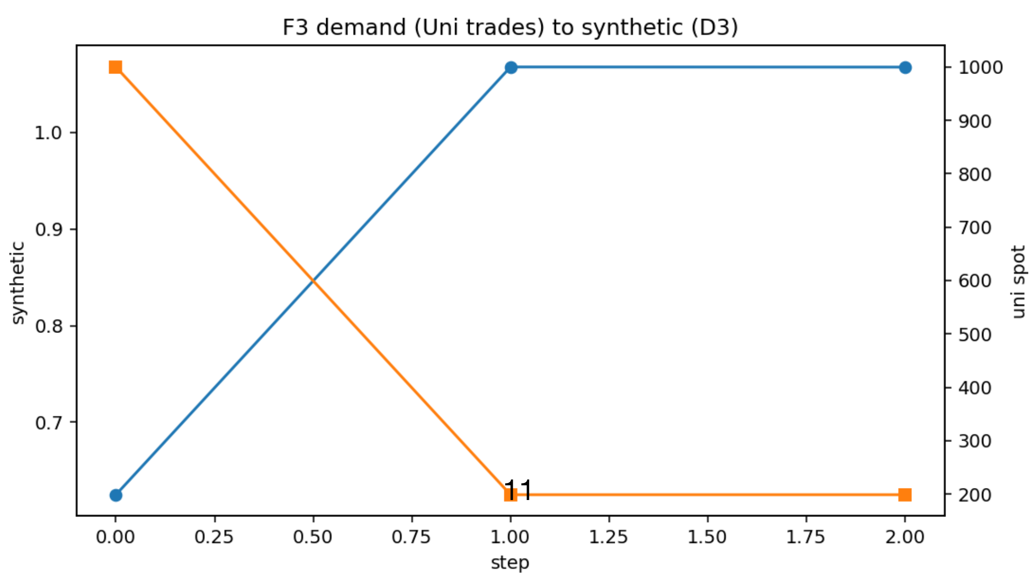


Figure 8: Fee as economic threshold: probes and fills under extreme volume.

ID	Result	One-line
D0	PASS	Inert Policy; defaults 1.05/0.95; mint blocked; warp inert
D1	PASS	First bond -> live
D2	PASS	Post-bond burn-side synth; mint reverts; no expansion on warp
D3	PASS	Mint-allowed after inventory path + Uni trades; preview = exec exact
D4	PASS	Burn-allowed; burn executes
D5	PASS	Open mint/burn live; no natural expansion over warp
D6	PASS	Serial capital mints increase supply (not expansion)
D7	PASS	Bonder reward path ≠ free-holder airdrop
D8	PASS	Policy expansion when rich; Open twin none
D9	PASS	Protocol compound raises protocol NFT BPT principal



8. Composition types

Four **live** families share the DETF pattern; only the reserve shape changes:

Type	Choose when
Single Standard Exchange	Exactly one SE vault + DETF weighted reserve
Multi-vault weighted	Multiple SE legs that must keep distinct valuations
Multi-vault stable	Like-kind rate targets
Mixed-buffer multi-vault stable	Shared buffer rate asset; burn buffer only

v1 empirics cover **Single SE only**. Other families are taxonomy and product law. Removed packages are out of research universe.

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9. Product hierarchy

Offer	Meaning
Premier	Create and deploy **your own DETFs** across package families
Protocol DETF	Same design class as a path to **share protocol fees** (amounts not guaranteed)

Neither is a registered securities fund. Fee sizes are not promises.

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10. Limitations and risks

1. **Hermetic only** — not mainnet TVL, APY, or reprice volume guarantees.
2. **Not a registered securities ETF** — no legal ownership of offchain underlyings.
3. **No peg, expansion APY, or claim coupon** — Policy bands and expansion are mechanisms, not guarantees.
4. **Open** removes price gates only — not fees, not returns, not expansion.
5. **IL and LVR** on underlying legs remain.
6. **Fee transferability** — SE research 5% fee is not every production pool.
7. **Post-bond synthetic** in hermetic Single SE may require inventory paths (not Uni alone) to re-enter mint-allowed.
8. **Protocol compound** single-sided join accepts self-leg weight skew in v1.
9. **Immutable instances** imply abandonment risk if misconfigured—not silent admin rescue.

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11. Conclusion

A DETF is a reproducible onchain share over a real reserve, with one pricing engine, explicit Policy or Open rules, bonding into protocol-owned depth, bond-ledger rewards, Policy-only natural expansion when rich, and keeper-free protocol compounding into reserve BPT. Nested SE rate providers keep marks honest under demand when wired on.

Hermetic research supports the lifecycle, gates, preview honesty, expansion negatives on Open, and protocol BPT growth on compound—**without inventing yield**.

Appendix A. Methodology

Item	Value
Profile	`FOUNDRY_PROFILE=default`
DETF runner	`./research/run_detf_single_se.sh`
DETF artifacts	`research/out/detf/singleSe/`
SE compare	`./research/run_rate_provider_compare.sh`
SE artifacts	`research/out/uniswapV2Se/rateProviderCompare/`
Telemetry	`series.jsonl`, stamped `meta.json`, `NOTES.md` per run
Production-first	CREATE3 facets; vault/DETF packages via manager registry; no mock SUT
Synthetic drive	Real Uni V2 trades (+ documented production inventory paths); no primary storage hacks

Companion materials: scenario FINDINGS, formal definitions, and figure manifest under `research/papers/detf-litepaper/` and `research/scenarios/detf/singleSe/`.

Appendix B. Claims map

ID	Claim	Status
C1	One share over reserve	Design
C2	Pricing engine = reserve pool	Design + measured
C3	Inert -> live via first bond	Measured D0–D1
C4	Policy / Open gates	Measured D2–D5
C5	Immutability after deploy	Product law
C6	Preview honesty	Measured D3 (exact)
C7	R+ residual approx 0	Measured SE
C8	Fee as arb threshold	Measured SE
C9	Bond vs mint books	Measured D7

ID	Claim	Status
C10	Four live composition types	Taxonomy
C11	Product hierarchy	Design
C12	Natural expansion Policy-only	Measured D5, D8
C13	Protocol compound -> BPT	Measured D9

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